

## SCHUEPFHEIM, Switzerland

WMO: 066510

Lat: 46.947N

Lon: 8.012E

Elev: 744

StdP: 92.70

Time Zone: 1.00 (EUC)

Period: 11-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF |
|---------------|------------|------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|-----|------|-----------------------|------|-----|
|               |            |      | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%  |      |                       |      |     |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |     |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p) |
| 2             | -12.4      | -9.2 | -14.9                         | 1.1 | -12.3 | -11.1 | 1.6 | -8.8 | 8.2                   | -0.3 | 7.4 | -0.1 | 2.9                   | 30   | N/A |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.7                   | 29.1            | 19.7 | 27.2 | 19.1 | 25.4 | 18.3 | 20.6                | 27.5 | 19.6 | 25.9 | 18.7 | 24.3 | 3.3                  | 50   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 18.2                            | 14.3 | 23.8 | 17.3 | 13.6 | 22.5 | 16.6 | 12.9 | 21.5 | 62.9          | 27.7 | 59.4 | 25.9 | 56.3 | 24.4 | 24.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 7.2           | 6.1  | 5.3 | DB                         | -15.3 | 31.5               | 3.1 | 1.5                                                | -17.6 | 32.6       | -19.4 | 33.5       | -21.1 | 34.3       | -23.4 | 35.4 | (4) |
| (5)               |      |     | WB                         | -16.1 | 22.1               | 2.9 | 1.0                                                | -18.2 | 22.8       | -19.9 | 23.5       | -21.5 | 24.0       | -23.6 | 24.8 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 8.7         | 0.2   | -0.2  | 4.3   | 8.0   | 11.4  | 16.0  | 17.7  | 17.3  | 13.5  | 9.5   | 4.4   | 1.6   | (6)  |  |
|                                                                        | DBStd       | 7.30        | 4.29  | 4.92  | 3.59  | 3.62  | 3.64  | 3.38  | 3.05  | 3.07  | 3.09  | 3.60  | 3.88  | 3.70  | (7)  |  |
|                                                                        | HDD10.0     | 1367        | 304   | 285   | 178   | 80    | 28    | 1     | 0     | 0     | 6     | 53    | 172   | 261   | (8)  |  |
|                                                                        | HDD18.3     | 3593        | 562   | 518   | 436   | 311   | 216   | 87    | 48    | 58    | 146   | 274   | 419   | 519   | (9)  |  |
|                                                                        | CDD10.0     | 886         | 1     | 0     | 1     | 19    | 71    | 179   | 239   | 226   | 111   | 37    | 3     | 0     | (10) |  |
|                                                                        | CDD18.3     | 72          | 0     | 0     | 0     | 0     | 1     | 16    | 28    | 26    | 1     | 0     | 0     | 0     | (11) |  |
|                                                                        | CDH23.3     | 865         | 0     | 0     | 0     | 2     | 17    | 217   | 337   | 268   | 25    | 1     | 0     | 0     | (12) |  |
|                                                                        | CDH26.7     | 198         | 0     | 0     | 0     | 0     | 1     | 57    | 85    | 55    | 0     | 0     | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 2.1         | 2.3   | 2.3   | 2.4   | 2.2   | 2.3   | 2.2   | 2.1   | 1.9   | 2.0   | 1.8   | 1.9   | 2.1   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 1561        | 95    | 99    | 105   | 118   | 166   | 171   | 178   | 172   | 123   | 106   | 112   | 115   | (15) |  |
|                                                                        | PrecMax     | 1951        | 240   | 284   | 238   | 234   | 277   | 270   | 342   | 393   | 240   | 182   | 262   | 273   | (16) |  |
|                                                                        | PrecMin     | 1208        | 7     | 32    | 43    | 33    | 39    | 91    | 71    | 41    | 54    | 5     | 1     | 5     | (17) |  |
|                                                                        | PrecStd     | 187         | 58    | 63    | 50    | 49    | 65    | 42    | 67    | 71    | 43    | 43    | 58    | 63    | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 12.2  | 14.5  | 17.8  | 22.6  | 25.4  | 30.9  | 31.6  | 30.2  | 25.6  | 21.2  | 16.7  | 11.7  | (19) |  |
|                                                                        |             | MCWB        | 8.9   | 7.9   | 10.2  | 14.0  | 16.8  | 20.5  | 20.4  | 20.2  | 18.2  | 14.8  | 11.2  | 7.8   | (20) |  |
|                                                                        | 2%          | DB          | 9.2   | 10.9  | 15.6  | 20.2  | 23.0  | 28.6  | 29.2  | 28.2  | 24.0  | 19.4  | 14.4  | 10.0  | (21) |  |
|                                                                        |             | MCWB        | 6.5   | 6.8   | 9.3   | 12.7  | 15.7  | 19.9  | 19.3  | 19.8  | 17.6  | 13.8  | 10.4  | 7.3   | (22) |  |
|                                                                        | 5%          | DB          | 7.5   | 8.4   | 13.6  | 18.5  | 21.0  | 25.9  | 27.1  | 26.4  | 21.9  | 17.2  | 12.0  | 8.8   | (23) |  |
|                                                                        |             | MCWB        | 5.4   | 5.5   | 8.3   | 11.8  | 14.3  | 18.4  | 18.6  | 19.3  | 16.5  | 13.0  | 9.2   | 6.5   | (24) |  |
|                                                                        | 10%         | DB          | 5.6   | 6.0   | 11.2  | 16.0  | 19.0  | 23.4  | 25.0  | 24.5  | 19.9  | 15.1  | 10.0  | 7.0   | (25) |  |
|                                                                        |             | MCWB        | 4.0   | 3.8   | 7.1   | 10.3  | 13.4  | 17.3  | 17.9  | 18.7  | 15.5  | 12.4  | 8.0   | 5.3   | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 9.3   | 8.6   | 11.0  | 14.5  | 17.7  | 22.3  | 21.1  | 21.6  | 19.2  | 16.1  | 12.0  | 9.2   | (27) |  |
|                                                                        |             | MCDB        | 11.7  | 11.9  | 16.0  | 21.0  | 24.2  | 29.5  | 29.4  | 27.6  | 23.6  | 19.8  | 15.7  | 10.0  | (28) |  |
|                                                                        | 2%          | WB          | 7.2   | 6.8   | 9.6   | 13.1  | 16.4  | 20.5  | 20.1  | 20.6  | 17.9  | 14.5  | 10.6  | 7.7   | (29) |  |
|                                                                        |             | MCDB        | 9.1   | 10.2  | 14.9  | 19.4  | 21.5  | 27.3  | 27.5  | 26.8  | 22.7  | 17.8  | 13.6  | 9.6   | (30) |  |
|                                                                        | 5%          | WB          | 5.5   | 5.3   | 8.5   | 12.0  | 14.8  | 19.1  | 19.1  | 19.6  | 17.0  | 13.6  | 9.5   | 6.7   | (31) |  |
|                                                                        |             | MCDB        | 7.2   | 8.0   | 13.2  | 17.4  | 18.9  | 25.2  | 25.8  | 25.4  | 21.2  | 16.3  | 11.5  | 8.6   | (32) |  |
|                                                                        | 10%         | WB          | 4.3   | 3.9   | 7.4   | 10.6  | 13.7  | 17.9  | 18.2  | 18.7  | 16.0  | 12.7  | 8.2   | 5.3   | (33) |  |
|                                                                        |             | MCDB        | 5.5   | 5.9   | 10.9  | 14.9  | 18.1  | 22.6  | 23.7  | 23.7  | 19.3  | 14.8  | 9.8   | 6.8   | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 6.5   | 7.6   | 9.7   | 10.8  | 10.2  | 10.5  | 10.7  | 10.8  | 9.8   | 8.9   | 7.4   | 7.2   | (35) |  |
|                                                                        |             | MCDBR       | 8.2   | 11.8  | 14.6  | 15.8  | 14.0  | 14.8  | 15.0  | 14.1  | 13.0  | 12.5  | 10.8  | 9.4   | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 6.1   | 7.7   | 9.2   | 9.0   | 7.3   | 7.3   | 6.7   | 6.9   | 7.1   | 7.7   | 7.1   | 6.5   | (37) |  |
|                                                                        |             | MCWBR       | 7.8   | 10.9  | 13.6  | 14.6  | 12.6  | 13.6  | 13.7  | 13.2  | 12.0  | 11.0  | 9.9   | 8.5   | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.279 | 0.308 | 0.351 | 0.391 | 0.408 | 0.411 | 0.413 | 0.401 | 0.374 | 0.346 | 0.309 | 0.277 | (40) |  |
|                                                                        |             | taud        | 2.385 | 2.314 | 2.261 | 2.212 | 2.224 | 2.264 | 2.256 | 2.301 | 2.360 | 2.415 | 2.445 | 2.404 | (41) |  |
|                                                                        | Edn at Noon |             | 816   | 858   | 866   | 861   | 858   | 855   | 848   | 843   | 834   | 804   | 773   | 775   | (42) |  |
|                                                                        |             | Edh at Noon | 70    | 93    | 114   | 132   | 136   | 131   | 131   | 120   | 103   | 84    | 67    | 61    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 1.32  | 2.21  | 3.40  | 4.38  | 4.94  | 5.80  | 5.67  | 4.93  | 3.80  | 2.44  | 1.39  | 1.09  | (44) |  |
|                                                                        | RadStd      |             | 0.16  | 0.30  | 0.49  | 0.66  | 0.47  | 0.53  | 0.50  | 0.46  | 0.36  | 0.26  | 0.19  | 0.17  | (45) |  |

## Historical Trends

|                               |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                               |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                               |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only             |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (121 neighbors) |  | N/A   | N/A     | N/A    | +0.73   | +0.40 | +0.31 | N/A         | N/A     | +58     | +16     |

Nomenclature: See separate page